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The Editors
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Dear Editors,

I submit the manuscript “A Coupled Chemo-Mechanical and Life-Cycle Framework for Oyster-Shell-Derived Cementitious Materials in Aggressive Infrastructure Environments” for consideration as a research article in Construction and Building Materials.

Oyster-shell powder is promoted as a low-carbon cement substitute, but the claimed joint carbon and durability benefit is usually asserted rather than tested. The manuscript asks, under stated uncertainty, when shell-derived concrete delivers a simultaneous carbon and service-life advantage over ordinary Portland cement concrete and where that claim fails. It separates uncalcined shell powder replacing cement (P1) from shell calcium carbonate replacing limestone in the kiln feed (P2) and couples hydration, porosity, chloride, acid and leaching transport, damage accumulation and cradle-to-gate carbon in paired Monte Carlo analyses of a pavement, a dam and a marine culvert, with Sobol sensitivity and an explicit falsification region.

The paper tests its own strength model against published data. The original inert-filler model is rejected by scallop-shell concrete data (Abbas and Kudukkan 2025): it predicts relative strengths of 0.79, 0.61 and 0.43 for three mixes against 1.00, 0.86 and 0.75 observed. An effective strength-efficiency coefficient calibrated on that study does not transfer to independent oyster-shell data: on 11 points from three studies its mean absolute error is 0.12, against 0.15 for the inert model and 0.14 for a dilution-only rule, and the coefficient differs between studies from -0.40 to 1.36. The primary results therefore use a pooled random-effects distribution that expresses this heterogeneity, and the original and single-study variants are reported as ablations.

The principal findings are conditional and largely unfavourable to the claim. At 25% replacement the probability of a carbon cut of at least 5% with non-inferior life is at most 0.21 for the pavement, 0.14 for the dam and 0.03 for the marine culvert; against a 50% slag blended cement it is zero, because that reference cuts carbon by about 42% on its own. Chloride transport remains uncalibrated, and a formulation in which the filler also refines the pores would raise the pavement probability to 0.45, so the durability conclusion is bracketed and not determined.

The conditions of every probability are collected in one table, a leave-one-study-out analysis of the pooled distribution shows that the qualitative conclusions do not depend on any one dataset (the

median pavement carbon cut moves between 9% and 20%), and service life is reported under fixed transport models spanning the favourable and unfavourable ends. The framework is presented as a computational uncertainty study, not a validated service-life model.

A companion manuscript prepared for the Journal of Cleaner Production treats the competing use of recovered shell, reef restoration; it takes results from this paper as inputs and is companion work, not independent validation of them.

Code, automated checks, all numerical results and the Supplementary Information are openly available at https://github.com/sandlerleon/oyster-shell-concrete-mc and archived at Zenodo (software, doi:10.5281/zenodo.23867048; manuscript preprint, doi:10.5281/zenodo.23867044). The original model was re-run from a clean directory before any change and its results reproduced to numerical precision. No new experiments are reported; the literature screen supporting the stated gap is limited and is described as such, and two of the independent datasets rest on assumptions listed in the manuscript.

A generative AI assistant (Claude, Anthropic) was used to help develop and test the simulation code, to run the analyses and to draft and revise the manuscript text and figures, as declared in the manuscript. I specified the research questions, reviewed and edited all content and take full responsibility for it. The manuscript is original, has not been published elsewhere and is not under consideration at any other journal. I am the sole author, I have no competing interests, and the work received no specific funding.

Thank you for considering this submission.

Sincerely,

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